Milestone 2: Dataset Preparation

Socratic AI Tutor

Group 2 | Data Science and AI Lab, IIT Madras BS

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What we prepared

Dataset	Used for	Size after preparation
ASSISTments 2009	Learner model (DKT)	282,619 answers, 3,852 students
MathDial	Socratic model fine-tuning	18,541 tutor replies
NCERT Maths + Science	Course pack (study text)	3,164 text chunks, 10 books

All three are public. Raw files are not committed; scripts in data/ rebuild everything.

ASSISTments: learner model data

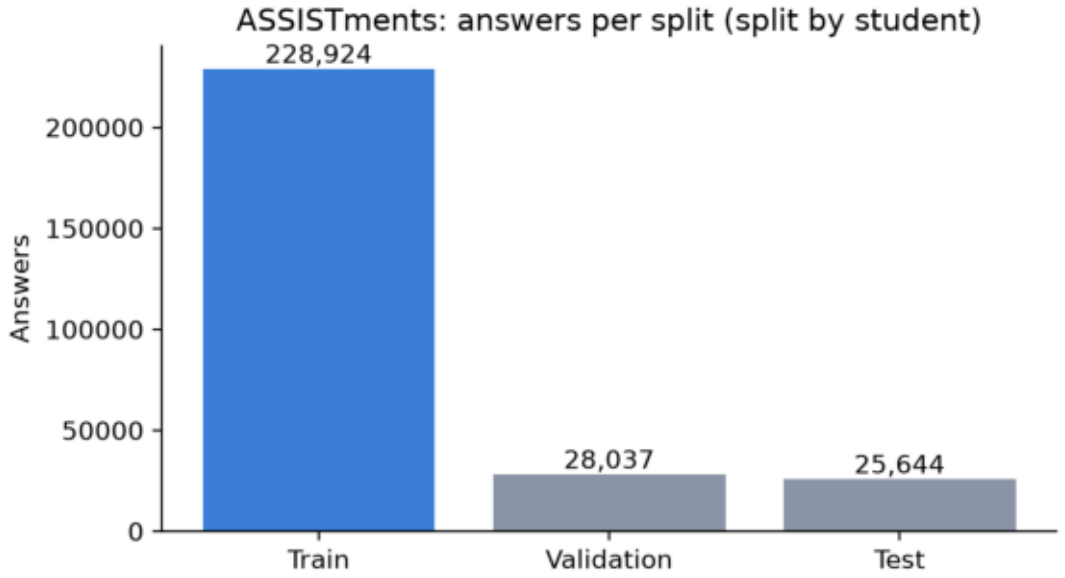
- Raw: 346,860 answers from 4,217 students

- Removed 63,755 answers with no skill label

- Dropped students with fewer than 3 answers (pyKT recipe)

- 149 skills, histories cut into windows of up to 200 answers

- Split by student: 80% train, 10% validation, 10% test



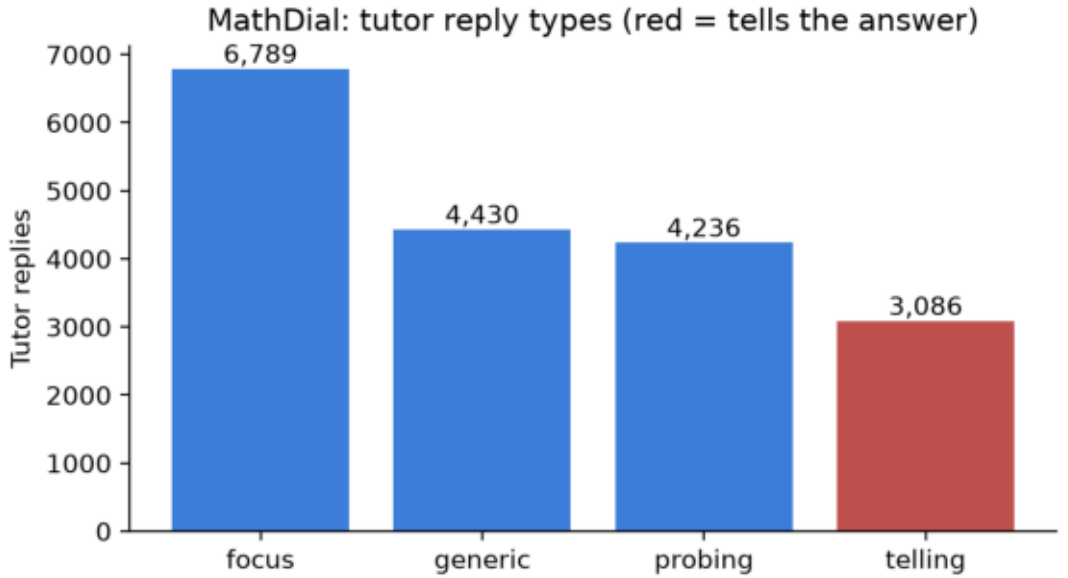
MathDial: Socratic model data

- One example per tutor reply: problem + conversation so far -> next reply

- Each reply keeps MathDial's label: probing, focus, generic, telling

- 'Telling' replies and replies containing the final answer are flagged and not used as training targets

- Validation split by problem, so a problem never sits in both train and validation



Version B: data prepared by Hariom Pandey

- A second preparation of ASSISTments and MathDial, kept so M3 can compare both versions

- DKT arrays: 82 skills, 3,408 students, 256,060 answers; one number per answer (skill + 82 x correct), padded to 200 with a mask; 80/20 split by student

- Chat corpus: 35,151 SocraTeach dialogues + 2,262 MathDial conversations = 37,413, for fine-tuning

- Differs from version A: 82 vs 149 skills, no validation set, no answer-leak flag; different source file

- Do not mix: B's MathDial conversations include version A's validation dialogues

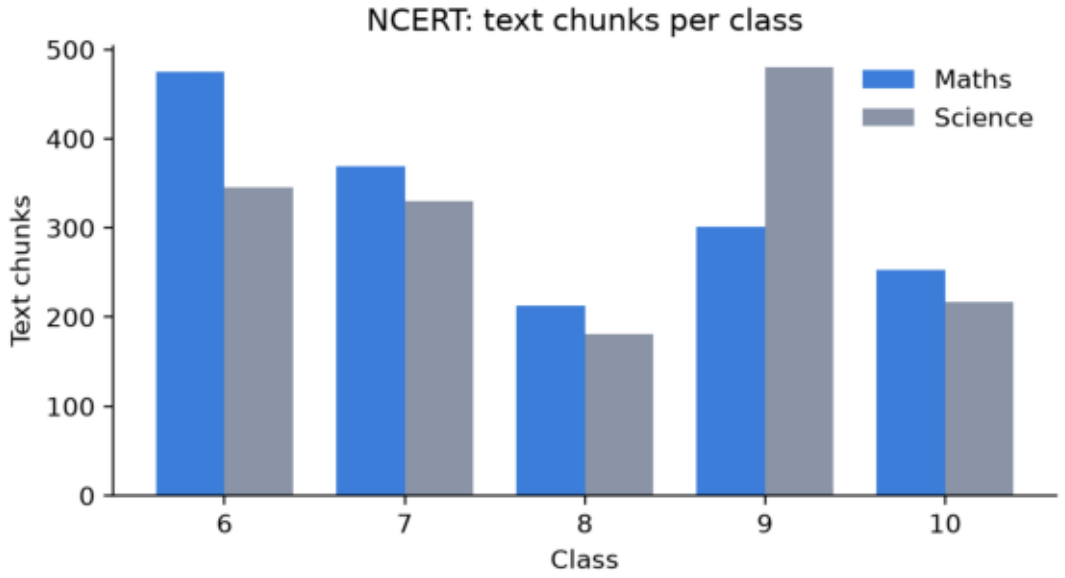
NCERT: study text

- Maths and Science, classes 6 to 10 (classes act as levels)

- 2,401 pages cleaned and cut into chunks of up to 250 words

- Each chunk keeps subject, class, chapter and page, so the tutor can point to the exact page

- Text stays local: the books are public but not ours to redistribute



Course packs: concepts and questions

- Hand-written chains of concepts that build across levels

- Example: percentages -> fractions -> multiplication and division

- Each concept links to its NCERT chapter, so the tutor can point to the page

- Practice questions have a hint and difficulty; answers are computed in code

Pack	Concepts	Prerequisite links	Questions
NCERT Maths	10	13	52
NCERT Science	6	6	18

Our own test scenarios

- Backtracking: 82 cases (correct, careless, hint, gap one or two levels down)

- Misconceptions: 32 wrong answers with the error behind each

- Guardrails: 47 messages (on-topic, off-topic, answer request)

- Each backtracking case carries the learner's mastery, so cases with the same inputs always have the same label

- Never used for training or tuning; a second member reviews every label

Sample records

```
ASSISTments A  {"user_id": 14, "skills": [136,136,136,144,137], "correct": [0,1,0,0,0]}
ASSISTments B  train,80959,chunk 0,step 1,skill 9,correct 1,x=91   (x = skill + 82 x correct)
MathDial       {"act": "probing", "keep": true, "leaks_answer": false,
                "target": "Steven, If you had 4 of something and tripled that amount..."}
Chat corpus B  {"source": "mathdial", "messages": [{"role": "system|assistant|user", ...}]}
Scenario       {"type": "careless_mistake", "mastery": {"concept": 0.9}, "expected_first_action": "probe"}
Guardrail      {"message": "What is the weather like today?", "label": "off_topic"}
```

More samples and plots: Milestone2_EDA.ipynb

Processed dataset: folders and splits

```
hosted_dataset/
  version_a_assistments/      train / val / test .jsonl, skill_map.csv
  version_b_assistments_dkt/  dkt_data.npz, dkt_processed_steps.csv
  mathdial/                   train / val / test .jsonl
  version_b_socratic_chat/    socratic_tutor_combined.jsonl
  ncert_metadata/             chapters.csv (text not shared)
  course_pack_specs/          specs.py
  test_scenarios/             backtracking, misconceptions, guardrails

Splits: ASSISTments A 80/10/10 by student | ASSISTments B 80/20 by student
        MathDial train / val (by problem) / MathDial's own test
```

Hosted dataset (Google Drive): https://drive.google.com/drive/folders/15vIRFghBsqfHjvHrIztgvVSUv2vP4tws

Checks and limits

- 26 automatic checks pass: no student or problem shared across splits, valid graphs, no hint contains its answer, consistent labels

- ASSISTments is US data and has no session information

- 283 of 394 MathDial test problems also appear in train (MathDial's own split)

- NCERT text: some digits lost in older books; study passages need a manual pass

Status against the Milestone 1 plan

Plan	Status	Note
NCERT as primary course	Done	text, chunks, 2 course packs
IITM BS pack	Not done	pipeline is course-agnostic; add later
Guardrail methods compared	Not done	eval set built (47); methods in M3
Controller / misconception baselines	Not started	moves to M3
Class 3-5 content (level 1)	Not done	only classes 6-10 processed

Next steps

- Second member reviews concepts, prerequisites and scenarios

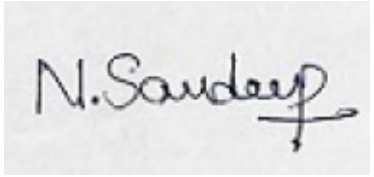
- Train DKT on both ASSISTments versions; published baseline AUC 0.754 (pyKT benchmark, Liu et al. 2022)

- Fine-tune the Socratic model on MathDial plus our NCERT examples

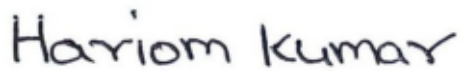
- Add the TA's test samples to our scenarios

Team sign-off

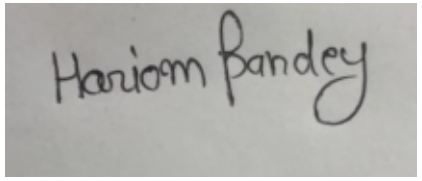
Sandeep Nalla (21f1006125)



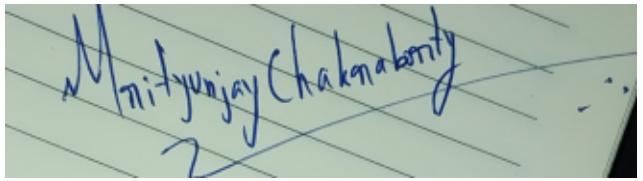
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Signed 29 October 2026